

Research Review Packet — DEMO

Billing Event Reconstruction and Payment-Delay Early-Warning Methods for Clinical Laboratory Settings

Run: run-20260922To40752Z-demo | Version 1.1 | Data mode: DEMO (synthetic) | This packet is an engineering review aid. It is not a filing, and it does not establish originality, attribution or eligibility.

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1. Core claims and limitations

- Module A reconstructs billing lineages with uncertainty preserved (precision 0.7642, recall 0.9895, refusal 0.0956 on hidden-truth adjacency).
- Module B predicts the first observable payment / no-payment close / still-open within 30 days; B1 improves snapshot log-loss 0.669 vs B0 0.736.
- C1 quality features add no incremental value (0.668); reported as-is.
- All results are synthetic-demo only; no real data, no external validation, no publication.

2. Method and rule summary

Linkage rules R1 (native explicit predecessor), R2 (same-bill-id version chains), R3 (amount/date-window weak evidence, uniqueness-gated). Amounts: gross / reversed / net observed / unallocated with over-allocation blocking. Prediction: discrete-time competing events, per-interval conditional probabilities with product-rule cumulative curves.

3. Interface (live captures)

Run

run-20260922T034730Z-demo ▾

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Delay early-warning (module B)

Snapshots predicted

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	snapshot_id	lineage_id	t	outcome	P_payment_l
0	run-CPL-R000-CPL-R00005@t20230722	run-CPL-R000-CPL-R00005	2023-07-22T01:00:27+00:00	still_open	0.022924
1	run-CPL-R000-CPL-R00009@t20230801	run-CPL-R000-CPL-R00009	2023-08-01T08:15:19+00:00	still_open	0.568177
2	run-CPL-R000-CPL-R00013@t20230809	run-CPL-R000-CPL-R00013	2023-08-09T18:38:57+00:00	still_open	0.521079
3	run-CPL-R000-CPL-R00024@t20230822	run-CPL-R000-CPL-R00024	2023-08-22T03:01:39+00:00	still_open	0.571602
4	run-CPL-R000-CPL-R00034@t20230907	run-CPL-R000-CPL-R00034	2023-09-07T03:48:55+00:00	payment	0.539430
5	run-CPL-R000-CPL-R00039@t20230922	run-CPL-R000-CPL-R00039	2023-09-22T02:52:15+00:00	still_open	0.025616
6	run-CPL-R000-CPL-R00040@t20230830	run-CPL-R000-CPL-R00040	2023-08-30T12:52:56+00:00	still_open	0.570209
7	run-CPL-R000-CPL-R00041@t20230828	run-CPL-R000-CPL-R00041	2023-08-28T14:33:03+00:00	payment	0.571525

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Event reconstruction

Lineages

1044

Links

123

Conflicts

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Lineage

run-CPL-R000-CPL-R00001 ▾

Status: OPEN

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SYNTHETIC DEMONSTRATION — NOT FOR FILING

Deploy

Manual review queue

Pending review items

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	run_id	review_type	record_ids	detail
0	run-20260922T034730Z-demo-A	MULTI_CANDIDATE_AMBIGUITY	CPL-R00113 CPL-R00401	same payer/service, amounts 7
1	run-20260922T034730Z-demo-A	MULTI_CANDIDATE_AMBIGUITY	CPL-R00113 CPL-R00402	same payer/service, amounts 7
2	run-20260922T034730Z-demo-A	MULTI_CANDIDATE_AMBIGUITY	CPL-R00336 CPL-R00457	same payer/service, amounts 8
3	run-20260922T034730Z-demo-A	MULTI_CANDIDATE_AMBIGUITY	CPL-R00430 CPL-R00431	same payer/service, amounts 9
4	run-20260922T034730Z-demo-A	MULTI_CANDIDATE_AMBIGUITY	CPL-R00431 CPL-R00457	same payer/service, amounts 8
5	run-20260922T034730Z-demo-A	MULTI_CANDIDATE_AMBIGUITY	CPL-R00155 CPL-R00390	same payer/service, amounts 8
6	run-20260922T034730Z-demo-A	MULTI_CANDIDATE_AMBIGUITY	CPL-R00389 CPL-R00390	same payer/service, amounts 7
7	run-20260922T034730Z-demo-A	MULTI_CANDIDATE_AMBIGUITY	TRI-R00506 TRI-R00507	same payer/service, amounts 9

Deploy

No runs found. Execute scripts/run_demo.py first.

4. Data and test summary

Synthetic generator seed 42; 1000 lineages; expected dirty cases intercepted: 40; acceptance suite T01-T25 (see outputs/qa/junit.xml).

5. Model comparison (locked test set)

Model	Status	snap log-loss	snap Brier
Bo_age_stage_baseline	FITTED	0.7357284660871263	0.5144259569702185
B1_discrete_multinomial	FITTED	0.6692132603100102	0.4618050188966311
C1_quality_augmented	FITTED	0.6682848894134452	0.4615831802291266

6. Source correspondence

SYNTHETIC DEMONSTRATION — NOT FOR FILING

Every number above is traceable to the run artifacts: module_b/forecast_metrics.json, module_b/validation_report.json, outputs/qa/junit.xml. References: research/prior_work.csv.

7. Outstanding items

- E5 external application: NOT_OBTAINED
- E6 academic impact (publication/citations): NOT_OBTAINED
- E7 expert interpretation: NOT_OBTAINED
- Real-data authorization: NOT_OBTAINED (governance/data_permissions.json)
- Applicant attribution review: pending